National Aeronautics and
Space Administration

NASA's
Open-Source Science Initiative

Supporting a more equitable, impactful, and efficient scientific future

Dr. Steven Crawford
Science Data Officer for Policy
Chief Science Data Office
NASA Headquarters

The United States White House announces

2023: A Year of Open Science

A multi-agency initiative across the US Federal Government to spark change and inspire open science engagement through events and activities that will advance adoption of open science. This includes the development of plans in response to the White House OSTP Memo on Ensuring Free, Immediate, and Equitable Access to Federally Funded Research.

✦CDC✦DOC✦DOE✦State✦DOT✦GSA✦NASA✦NEH✦
✦NIST✦NOAA✦NSF✦Smithsonian✦USDA✦USGS✦

To date:

- ✦ White House recognizes 2023 as a Year of Open Science
- ✦ 15 US Agencies join: representing >$90B in science funding
- ✦ White House Fact Sheet
- ✦ Federal definition of open science
- ✦ 4 goals for A Year of Open Science
- ✦ Website: https://open.science.gov/

NASA’s method to put Open Science into practice

Policy and Governance

Core Data and Computing Services

NASA’s Open-Source Science Initiative

Incentives

Community

TOPS
NASA

Open Science

is the principle and practice of making research products and processes available to all, while respecting diverse cultures, maintaining security and privacy, and fostering collaborations, reproducibility and equity.

SPD-41a is SMD's updated Scientific Information Policy

SPD-41a updates the previously released SPD-41, which consolidated existing Federal and NASA policy on sharing scientific information.

Policy updates were developed with:

- SMD community input via workshops, 61 RFIs, and Townhalls
- National Academies studies
- White House OSTP Memo on Ensuring Free, Immediate, and Equitable Access to Federally Funded Research
- Presentations to over >1000 stakeholders since Dec 2022
- Open Source Science Guidance for Researchers and FAQ
- Each division has also released policies and templates for their communities.

Scientific Information Policy Website

https://science.nasa.gov/researchers/science-data/science-information-policy

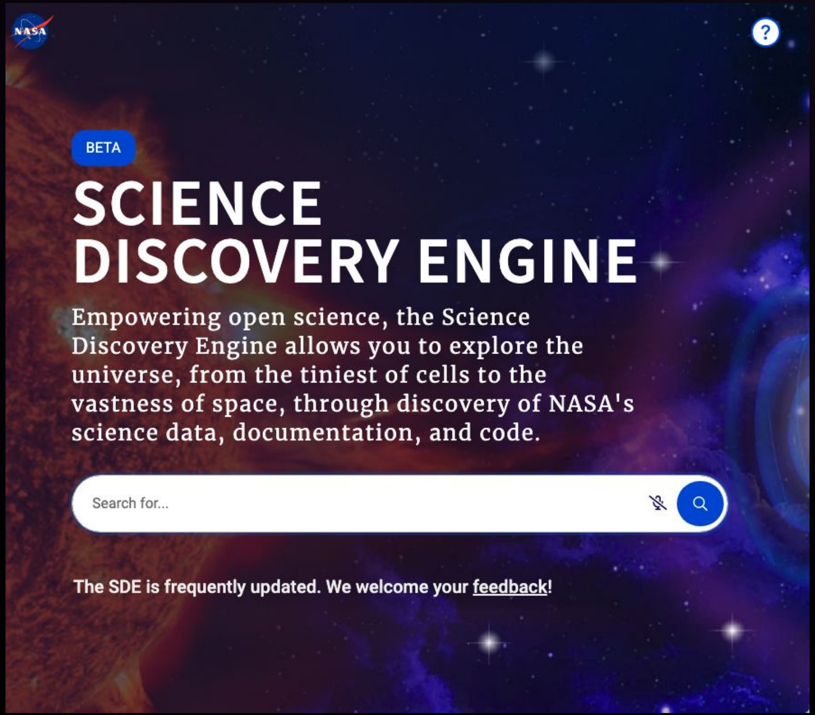
SPD-41a is SMD's updated Scientific Information Policy

SPD-41a is *forward looking* and will apply to all future SMD-funded scientific activities

Major Policy Updates

- Peer-reviewed publications are made openly available with no embargo period.
- Research data and software are shared at the time of publication or the end of the funding award.
- Mission data are released as soon as possible and unrestricted mission software is developed openly.
- Science workshops and meetings are held openly to enable broad participation.

Core Data and Computing Services Program



Science Discovery Engine

SMD data catalog to support discovery and access to complex scientific data across Divisions

Science Explorer

Extend the primary digital library portal for researchers in astrophysics, planetary science & heliophysics, the Astrophysics Data System (ADS), to support Earth and Biological and Physical Sciences

Data and Computing Infrastructure

A layered architecture on which SMD science Divisions can seamlessly and efficiently integrate their discipline-specific services such as data archives.

NASA is Sustaining Open-Source Software

NASA provides funding for open science through a solicitations and opportunities. This includes supporting innovative open science projects and open source tools, frameworks, and libraries.

NumPy

pandas

sunpy

julia

DEDALUS PROJECT

scikit learn

geemap

HDF

GDAL

NASA's Transform to Open Science (TOPS)

A 5-year mission to accelerate adoption of open science

Objectives:

- Increase understanding and adoption of open science principles and techniques
- Broaden participation by historically excluded communities
- Accelerate major scientific discoveries

Engagement **Resources** **Incentives**

Coordination

Learn more at: https://nasa.github.io/Transform-to-Open-Science/

TOPS Capacity Sharing: Open Science 101

5 Modules designed to introduce Open Science

ETHOS OF OPEN SCIENCE — What is open science? Why should I do it? How should I do it?

Open Science

OPEN TOOLS & RESOURCES — How to use popular open science tools

Open Tools

OPEN SOFTWARE — How to use, make, & share software

Open Software

OPEN DATA — How to use, make, & share open data

Open Data

OPEN RESULTS — Best practices for sharing all results and analysis, as well as peer reviewing

Open Results

Earn Badges at Each Level

TOPS

NASA

Complete All 5 & earn NASA Open Science Badge

The NASA Open Science Certificate indicates researchers have key open science skills

- Able to use digital tools to perform open science (e.g., ORCID, Zenodo, Github accounts)
- Familiar with data management and software management plan best practices and resources
- Grow connections across a community of open science practitioners

Enroll to learn more!

*A **community developed** introduction to open science with inclusivity, accessibility, and diversity at the forefront.*